Compiler Construction

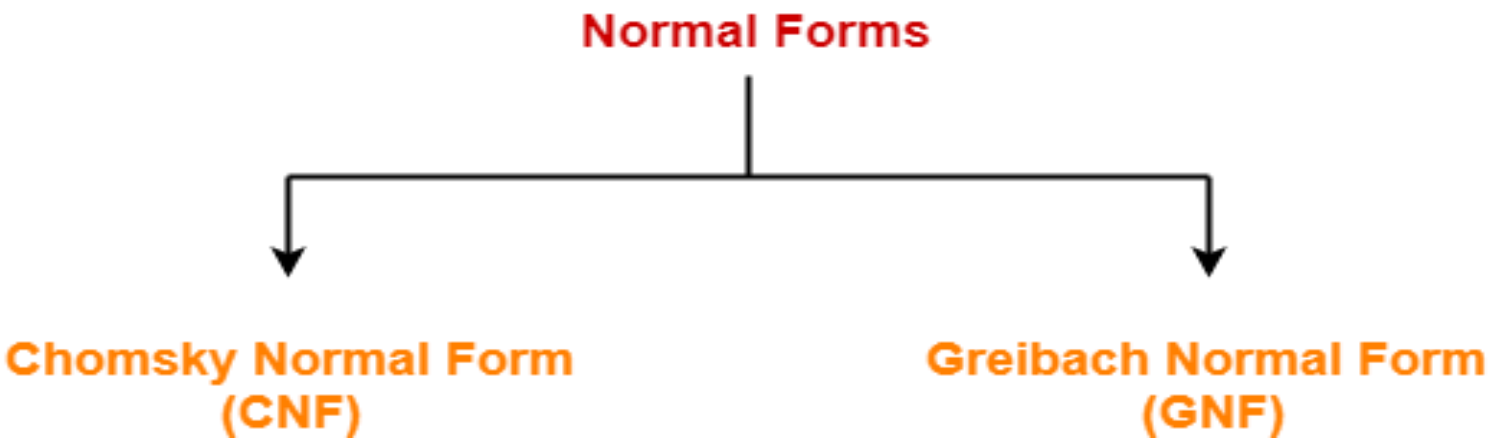
Chapter 5(Normal Forms)

Dr. Doaa Shebl
Faculty of Computers and Artificial Intelligence
Beni-Suef University

Chapter 5- Normal Forms

Types of Normal Forms

The most frequently used normal forms are



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Restriction for Normal Form

* nonrecursive start symbol.
* **Elimination of Lambda Rules**
* **Elimination of Chain Rules**
* **Useless Symbols**

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nonrecursive start symbol.

Example:

G: $S \to aS \mid AB \mid AC$

$A \to aA \mid \lambda$

$B \to bB \mid bS$

$C \to cC \mid \lambda$

G′: $S' \to S$

$S \to aS \mid AB \mid AC$

$A \to aA \mid \lambda$

$B \to bB \mid bS$

$C \to cC \mid \lambda$

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Elimination of Lambda Rules

Example:

$$S \rightarrow SaB \mid aB$$
$$B \rightarrow bB \mid \lambda.$$

$$\Rightarrow$$

$$S \rightarrow SaB \mid Sa \mid aB \mid a$$
$$B \rightarrow bB \mid b$$

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Elimination of Chain Rules

$$A \rightarrow aA \mid a \mid B$$

$$B \rightarrow bB \mid b$$

$$A \rightarrow aA \mid a \mid B$$

$$B \rightarrow bB \mid b$$

$$A \rightarrow aA \mid a \mid bB \mid b$$

$$B \rightarrow bB \mid b$$

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Useless Symbols

G: $S \to AC \mid BS \mid B$

$A \to aA \mid aF$

$B \to CF \mid b$

$C \to cC \mid D$

$D \to aD \mid BD \mid C$

$E \to aA \mid BSA$

$F \to bB \mid b.$

G: $S \to AC \mid BS \mid B$

$A \to aA \mid aF$

$B \to CF \mid b$

$C \to cC \mid D$

$D \to aD \mid BD \mid C$

$E \to aA \mid BSA$

$F \to bB \mid b.$

Iteration	TERM	PREV
0	$\{B, F\}$	
1	$\{B, F, A, S\}$	$\{B, F\}$

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G: $S \rightarrow AC \mid BS \mid B$

$A \rightarrow aA \mid aF$

$B \rightarrow CF \mid b$

$C \rightarrow cC \mid D$

$D \rightarrow aD \mid BD \mid C$

$E \rightarrow aA \mid BSA$

$F \rightarrow bB \mid b.$

Iteration	TERM	PREV
0	$\{B, F\}$	
1	$\{B, F, A, S\}$	$\{B, F\}$
2	$\{B, F, A, S, E\}$	$\{B, F, A, S\}$

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Iteration	TERM	PREV
0	$\{B, F\}$	
1	$\{B, F, A, S\}$	$\{B, F\}$
2	$\{B, F, A, S, E\}$	$\{B, F, A, S\}$
3	$\{B, F, A, S, E\}$	$\{B, F, A, S, E\}$

G: $S \to AC \mid BS \mid B$

$A \to aA \mid aF$

$B \to CF \mid b$

$C \to cC \mid D$

$D \to aD \mid BD \mid C$

$E \to aA \mid BSA$

$F \to bB \mid b.$

$\Rightarrow$

$S \to BS \mid B$

$A \to aA \mid aF$

$B \to b$

$E \to aA \mid BSA$

$F \to bB \mid b.$

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Chomsky Normal Form

From here, we infer- To be in CNF, all the productions must derive either two non-terminals or a single terminal.

CNF restricts the number of symbols on the right side of a production to be two.

The two symbols must be non-terminals or a single terminal.

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Chomsky Normal Form

A context free grammar is said to be in chomsky normal form (CNF) if all its productions are of the form-

A → **BC** or **A** → **a**

where A, B, C are non-terminals and a is a terminal

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<u>Example-</u>

$S \rightarrow AB$
$A \rightarrow a$
$B \rightarrow b$

This context free grammar is in chomsky normal form.

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Example1 :

Convert the following grammar to <u>Chomsky normal form</u>

$G: S \rightarrow aAB \,/\, a\,A \,/\, b\,B$
$A \rightarrow a\,Ab \,/\, aB \,/\, a$
$B \rightarrow bA/\ b$

Solution

Step 1 :

The productions already in chomsky normal form are:

$A \rightarrow a$
$B \rightarrow b$

These productions will remain as they are.

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Step 2 :

The productions not in chomsky normal form are-

$G: S \rightarrow aAB \,/\, a\,A \,/\, b\,B$
$A \rightarrow a\,Ab \,/\, aB$
$B \rightarrow bA$

We will convert these productions in chomsky normal form

Step 3:
Replace the terminal symbols <u>a</u> and <u>b</u> by new variables T_1 and T_2

This is done by introducing the following two new productions in the grammar:
$T_1 \rightarrow a$
$T_2 \rightarrow b$

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$G: S \rightarrow aAB \,/\, aA \,/\, bB$
$A \rightarrow aAb \,/\, aB \,/\, a$
$B \rightarrow bA \,/\, b$

Now, the productions $S \rightarrow aA$ modifies to $S \rightarrow T_1 A$

the productions $S \rightarrow bB$ modifies to $S \rightarrow T_2 B$

the productions $A \rightarrow aB$ modifies to $A \rightarrow T_1 B$

the productions $B \rightarrow bA$ modifies to $B \rightarrow T_2 A$

the productions $A \rightarrow aAb$ modifies to $A \rightarrow aAT_2$

$G: S \rightarrow aAB \,/\, T_1 A \,/\, T_2 B$
$A \rightarrow aAT_2 \,/\, T_1 B \,/\, a$
$B \rightarrow T_2 A \,/\, b$
$T_1 \rightarrow a$
$T_2 \rightarrow b$

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$G: S \rightarrow aAB \,/\, T_1 A \,/\, T_2 B$
$A \rightarrow a\,AT_2 \,/\, T_1 B \;\; / \;\; a$
$B \rightarrow T_2 A \,/\; b$
$T_1 \rightarrow a$
$T_2 \rightarrow b$

Step 3:

Replace AB and AT_2 by new variables T_3 and T_4 respectively.

This is done by introducing the following two new productions in the grammar:

$T_3 \rightarrow AB$
$T_4 \rightarrow AT_2$

Now, the productions $S \rightarrow aAB$, and $A \rightarrow a\,AT2$ modifies to:
$S \rightarrow T_1\, T_3$
$A \rightarrow T_1\, T_4$

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Finally:

$G: S \rightarrow aAB \,/\, a\,A \,/\, b\,B$
$A \rightarrow a\,Ab \,/\, aB \;\; / \, a$
$B \rightarrow bA / \;\; b$

<u>Chomsky normal form:</u>

$G: S \rightarrow T_1\, T_3 \,/\, T_1\, A \,/\, T_2\, B$
$A \rightarrow A \rightarrow T_1\, T_4 / \, T_1\, B \;\; / \;\; a$
$B \rightarrow T_2\, A \,/ \;\; b$
$T_1 \rightarrow a$
$T_2 \rightarrow b$
$T_3 \rightarrow AB$
$T_4 \rightarrow AT_2$

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Greibach Normal Form

A context-free grammar $G = (V, \Sigma, P, S)$ is in **Greibach normal form** if each rule has one of the following forms:

i) $A \to aA_1A_2 \ldots A_n$

ii) $A \to a$

iii) $S \to \lambda$,

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Example 2: Convert the following grammar to <u>Greibach normal form</u>

$G: S \rightarrow aAB \,/\, a\,A \,/\, b\,B$
$A \rightarrow a\,Ab \,/\, aB \,/\, a$
$B \rightarrow bA/\ b$

Solution

Step 1 :

The productions already in Greibach normal form are:

$G: S \rightarrow aAB \,/\, a\,A \,/\, b\,B$
$A \rightarrow aB \,/\, a$
$B \rightarrow bA/\ b$

These productions will remain as they are

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Step 2 :

The productions not in Greibach normal form are-

$A \rightarrow a\,Ab$

We will convert this production in Greibach normal form

Step 3:

Replace the terminal symbol <u>b</u> by new variables T_1

$T_1 \rightarrow b$

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Finally :

Greibach normal form

$G: S \rightarrow aAB \,/\, a\,A \,/\, b\,B$

$A \rightarrow a\,AT_1 \,/\, aB \,/\, a$

$B \rightarrow bA/\ b$

$T_1 \rightarrow b$

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Example 3:
Convert the following grammar to <u>Greibach normal form</u>

$G: S \rightarrow aAB \,/\, a\,A \,/\, B\,b$
$A \rightarrow a\,Ab \,/\, aB \,/\, a$
$B \rightarrow bA/\ b$

Solution

Step 1 :

The productions already in Greibach normal form are:

$G: S \rightarrow aAB \,/\, a\,A$
$A \rightarrow aB \,/\, a$
$B \rightarrow bA/\ b$

These productions will remain as they are

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Step 2 :

The productions not in Greibach normal form are-

$S \rightarrow B\ b$
$A \rightarrow a\ Ab$

We will convert this production in Greibach normal form

Step 3:
Replace the terminal symbol <u>b</u> by new variables T_1

$T_1 \rightarrow b$

$S \rightarrow B\ b$ *?????*

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$G: S \rightarrow aAB \,/\, a\,A \,/\, B\,b$
$A \rightarrow a\,AT_1 \,/\, aB \,/\, a$
$B \rightarrow bA/\ b$
$T_1 \rightarrow \mathbf{b}$

$S \rightarrow B\ \ b$

$B \rightarrow bA/\ b$

$S \rightarrow bA\,b \,/\, b\,b$

$G: S \rightarrow aAB \,/\, a\,A \,/\, bA\,b \,/\, bb$
$A \rightarrow a\,AT_1 \,/\, aB \,/\, a$
$B \rightarrow bA/\ b$
$T_1 \rightarrow \mathbf{b}$

$G: S \rightarrow aAB \,/\, a\,A \,/\, bA\,T_1 \,/\, bT_1$
$A \rightarrow a\,AT_1 \,/\, aB \,/\, a$
$B \rightarrow bA/\ b$
$T_1 \rightarrow \mathbf{b}$